

Gender, Ideology, and Fiscal Policy

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Abstract

This paper studies whether women Finance Ministers produce different fiscal outcomes than men, and whether any such differences are because of their partisan ideological beliefs. I use a country-year panel covering Finance Ministers in over 170 countries from 1990 to 2023. I estimate the IV-TWFE model to test my hypothesis. The analysis identifies 391 women Finance Ministers in the sample post-1990. One main issue I came across is the 'glass cliff' phenomenon, that is, women are disproportionately appointed, especially during periods of crisis or tension, which generates downward bias in the estimates. I address this using two instruments, the count of women in other cabinet positions relative to the country's region-decade peer average and the Finance Minister's age at appointment. IV estimates indicate that women Finance Ministers improve the fiscal balance relative to men but there is no significant effect of their ideological beliefs. The ideological distribution of women Finance Ministers skews modestly left, which might indicate that there is a partisan selection rather than a gendered effect. But, the findings here show that the ideology coefficient is statistically indistinguishable from zero in all fiscal outcome specifications, indicating that the effect is driven by gender rather than ideology.

JEL Codes: E62, H62, J16, D72

1. Introduction

The representation of women in the cabinet and economic policymaking is quite low in the whole world. As of 2026, women were holding only 22.4% of cabinet member seats across the world (a reduction from 2024). And, the Finance Ministry, which is a significant cabinet position, is one of those ministries which have been dominated by men. In this context, one crucial question arises here, how is the representation and functionality of women in the Finance Ministry? and how is it different from that of men?

This paper tries to answer these questions based on country-year panel data of over 170 (roughly 173) countries from 1990 to 2023. The dataset WhoGov (*Nyrup and Bramwell, 2020*) has individual cabinet ministerial information from most countries in the world, which is combined with GLID (Global Leader Ideology Dataset), and IMF World Economic Outlook (WEO) to conduct the analysis. The party ideology of the individual Finance Ministers has been manually coded using AI (Claude and GPT-4) and validated against the GLID's ideology variable for Head of the Government and the Wikipedia pages of each party from each country on a scale of 1-6. I finally have a dataset with only 411 female Finance Ministers since 1965 till now.

A central empirical challenge I come across in this paper is endogeneity and the selective hiring of women in cabinet ministerial positions ("glass Cliff phenomenon"). Finance Ministers are not randomly elected or selected, they are chosen by the Heads of the Government. This selection is often done during periods of tension and crisis. A large literature on the "glass cliff" phenomenon describes how women are disproportionately selected into leadership during crises (*Armstrong et al.; Ryan and Haslam, 2024; 2005*). In the fiscal context, *Armstrong et al. (2024)* demonstrates that financial crises significantly raises the possibility of a women getting appointed as Finance Minister, while *Kern, Reinsberg, and Romelli (2025)* document a parallel dynamic in central banking appointments as well. This selection mechanism, quite naturally, generates a negative bias in the estimates, since women face (rather more like they

are asked to survive) deteriorating fiscal conditions more often than men.

I have mainly used two instruments here. First is the variation in cabinet gender representation across countries in different regions, proxied by the number of women in non-Finance ministry positions compared to the regional-decade average. Second is the age of Finance Ministers at time of their appointment. This second instrument was chosen based on the skewness in women’s career paths to reach higher political office and the experience or exposure they get to reach that position in their careers (most of them are able to reach that position at a mature age).

The findings of my study indicate that female finance ministers enhance fiscal performance of their countries, with the impact size rising from insignificant in TWFE specification under just-identified IV and overidentified IV. It is almost echoing the glass cliff literature I was talking about just above, where the estimated impacts follow a monotonically rising path with respect to the degree of exogeneity of the instruments used in the analysis. The overidentification approach suggests an impact roughly equal to two-and-a-half standard deviations of fiscal balance within the countries (*Alesina, Favero, and Giavazzi, 2019*).

The second part of this paper is about these women’s ideological partisanship and its impact on their fiscal policy choices. Female Finance Ministers exhibit slightly more leftist views compared to their male counterparts, as seen in my data. But in this regard, the obvious concern would be whether the observed fiscal gains from having a female finance minister can be attributed to the leftist administrations or the gender of the finance minister. I find that the ideology of finance ministers does not predict their fiscal choices, irrespective of which country or system they are in. Moreover, the interaction between gender and ideology exhibits no statistically significant association with any of the three fiscal variables used in my model.

This paper offers five unique contributions. **First**, it offers the most comprehensive cross-country panel test yet of the causal impact of female Finance Ministers on fiscal performance, spanning 173 countries over thirty years. **Second**, it incorporates corrections

for glass-cliff selection bias through two new instruments, providing causal estimates. **Third**, it rejects the idea that partisan ideology has an influence on the decisions regarding the fiscal policy choices. **Fourth**, it uses instruments that use gender norms, which are widespread across different countries and career-queue processes that might be useful and meaningful. **Fifth**, it generates innovative data on ideology at the finance ministry level, including finance ministers from 170 countries between 1990 and 2023.

The paper is organized as follows. After an overview of the literature in the following Section 2, Section 3 covers the data part. Identification strategy is described in Section 4, while the results can be found in Section 5 and the robustness checks in Section 6. Lastly conclusion is in Section 7.

2. Background and Existing Evidence

This paper draws on several related literatures like political economy of fiscal policy, gender and political leadership, glass cliff and appointment in times of crisis, and gender and economic institutions. Across all of these literatures, an iterative idea comes up that the results of fiscal policy depend not only on the macroeconomic situation, but also on political institutions, policymakers, and the conditions under which they come to power in the country under consideration.

One section of the literature used in this paper illustrates how institutions and electoral incentives affect the fiscal outcomes. *Alesina and Perotti* (1996) have shown that political instability and fragmentation are often associated with higher fiscal deficits, whereas *Persson and Tabellini* (2002) have demonstrated that differences are due to the election system and regime type, which influences the amount and composition of government expenditures. Even if an institutional structure in question is supposed to be independent, electoral incentives may drive its decisions to a great extent, directly or indirectly (*Beck*, 1987). For the individual decision-makers, *Moessinger* (2012) have found that personal characteristics of Finance Ministers matter quite a lot for their fiscal policies.

Party affiliation and ideology impact the policies in different cabinet positions, which is a very common finding of many papers in this area. *Hibbs* (1977) suggests left-leaning regimes adopt more expansive fiscal policies compared to those with conservative regimes. This has also been replicated using the democratic regimes by *Potrafke* (2011). Comparable results can be observed in the case of monetary authorities as well, as seen in *van Ommersen and Piccillo* (2021). Given these insights, I believe including the finance minister’s ideology as a control variable and possible moderator is crucial to find out its impact, if there is any.

A good amount of literature shows how women leaders differ from men in their leadership positions in different aspects. Utilizing gender quota rotations in the Indian villages, *Chattopadhyay and Duflo* (2004) find that women leaders spend on goods favoured by women and children, whereas *Pande* (2003) show that descriptive representation influences the policymaking process regardless of which party the candidate belongs to. There is some contradictory evidence for economic organizations at an upper level as well. For instance, *Masciandaro, Profeta, and Romelli* (2020) argues that higher numbers of women in monetary policy committees in the central bank lead to more adoption of hawkish policies regarding interest rates, while ? (?) find that females in Federal Open Market Committee (FOMC) behave in ways quite different from their male colleagues and influence financial markets. Moreover, *Suzuki and Avellaneda* (2018) confirms the lower risk of fiscal financial loss among women public officials in the Japanese local governments.

Although a fair amount of literature exists for each of these topics, it is rare to find any information pertaining to the effect that female Finance Ministers have on fiscal policy outcomes. The current literature focuses heavily on the factors that lead to appointment as opposed to policy effects. According to *Krook and O’Brien* (2012), *O’Brien et al.* (2015), and *Field* (2021), left-leaning political parties, parliaments, and gender quotas all affect women’s presence within positions of power, while women are provided with less prestigious ministries. Yet, no articles focus on the economic effect of women in such roles.

There is an evident problem with endogenous selection into office during crises. This

issue relates to the literature on the “glass cliff,” which originated from *Ryan and Haslam* (2005), who found that women were more frequently appointed to leadership roles in declining organisations. From a political economy perspective, *Armstrong et al.* (2024) found that financial crises raise the likelihood of appointing women Finance Ministers, especially in countries where women have not held executive positions before. As a result, any naïve estimates of the link between women Finance Ministers and fiscal outcomes will be downward biased due to endogenous selection.

The same trends are observed in monetary institutions. *Kern, Reinsberg, and Romelli* (2025) find that sovereign debt crises positively predict the hiring of women in central bank board positions and that female board members are more likely to conduct conservative monetary policy. The causation might be, however, due either to preference differences or to the constraints under which women operate once appointed to these positions. Understanding the difference between these two factors helps explain the instrumental variable approach used in this paper.

Lastly, there is an emerging body of literature looking at the gender-economics-institution-policy-making area. In particular, *Bodea and Kerner* (2025) finds that people tend to trust female central bank governors much less compared to their male colleagues when inflation rates are particularly high, whereas *McDowell and Steinberg* (2024) show that the gender of an economic policymaker affects the perceived legitimacy of economic institutions. Last but not least, *Flamini et al.* (2023) show that macroeconomic policies definitely have gendered consequences, that is, the gender of the policymaker matters. In this context, this paper offers new insights by showing how women finance ministers affect fiscal policy outcomes.

3. Data

3.1. Dataset Construction

The analysis uses three primary datasets merged into a one country-year panel spanning the years between 1990 and 2023.

Cabinet minister data: WhoGov (*Nyrup and Bramwell, 2020*) provides individual-level on cabinet ministers across 177 countries from 1963 to 2023. The main variables include the minister’s name, gender, party affiliation, portfolio assignment, tenure dates, start and end dates, and history of other ministerial positions held, if that’s the case, among many other such detailed information. The merged panel contains 9,101 country-year observations, restricted to 1990–2023 for data coverage reasons, mainly dependent on the data for fiscal outcomes (WEO).

Ideology data: Finance Minister party ideology is coded on a 1–6 left-right scale (1 = far left, 5 = far right), with a separate category for independents/ technocrats, using AI-assisted (Claude and GPT-4) coding using Wikipedia biographical pages for each party in each country in their official language and in English. The The resulting minister-level variable is cross-checked against the head-of-government ideology variable from the GLID (*Brast et al., 2023*) dataset. I found that the agreement between these two sources exceeds 87% at the country-year level, the remaining discrepancies were resolved manually using party platform documents and Wikipedia page entries in the party’s official language and in English language. Ideology coverage is 95% across all Finance Ministers and 85% among women Finance Ministers, with missing observations concentrated in 2021–2023 due to GLID coding lag for recent governments or unavailable information for webpages of certain parties.

Table 5 reports the ideological distribution of finance ministers by their gender. Women finance ministers score 3.70 on the 1–6 scale on average, compared with 3.97 for men

($p = 0.002$), which means that women finance ministers are appointed by left-leaning parties more often than right-leaning parties. The gap is statistically significant but substantively modest, and does not map cleanly onto the distinct fiscal policy preferences, a point that I confirmed empirically in Section 5. The ideology variable covers all 9,101 observations.

Fiscal outcomes: Fiscal data is taken from the IMF World Economic Outlook (WEO) database, providing annual general government revenue, expenditure, fiscal balance, primary balance, and gross debt as shares of GDP for 196 countries from 1980 to 2023. I categorized the data into three main outcomes, the primary balance, fiscal balance, and the structural balance. WEO is preferred as a point of start because of its concise information on the fiscal side of this analysis. Later, I intend to use the IMF Government Finance Statistics for its broader country-year coverage and more detailed fiscal variables and their sub-categories.

Merged sample: The analysis is restricted to 1990–2023 to cover the post-Cold War period in which cabinet gender data are most reliable from all parts of the world. After merging and dropping observations with missing fiscal data, the core sample comprises 3,938–5,048 observations across the 173 countries. Table 1 reports full-sample summary statistics of the sample used to conduct further analysis, and Table 2 disaggregates key variables by the finance minister’s gender.

3.2. Treatment Variable

The primary treatment variable is:

$$\text{female_fm}_{it} = \mathbf{1}[\text{m_Finance}_{it} = 1 \cap \text{gender}_{it} = \text{Female}]. \quad (1)$$

The dataset identifies 391 women finance minister country-year observations across the 173 countries, representing 6.8% of the analysis sample.¹ Table 3 reports the distribution

¹Following WhoGov conventions, the observation unit is the country-year; a country with two Finance Ministers in the same year (e.g., due to a mid-year reshuffle) contributes two records. The indicator `female_fm`

across UN regional groupings. The treatment rate is highest in Latin America region and the Caribbean (8.7%) and lowest in the Middle East and North Africa (0.5%); the latter region is excluded from the main IV specifications as a sensitivity check. Table 4 documents a strongly positive secular trend, rising from under 2% of appointments in the 1990s to over 12% in 2020–2023.

4. Identification Strategy

4.1. The Glass Cliff Problem

Women’s finance minister appointments are not quite random with respect to prevailing fiscal conditions, as we might expect. The glass cliff literature documents that women are disproportionately elevated to senior leadership especially during periods of institutional distress and tension (*Cook and Glass; Ryan and Haslam; ?*, 2010; 2005; ?). In the fiscal context, this implies:

$$\text{Cov}(D_{it}, \varepsilon_{it} \mid \alpha_i, \lambda_t) < 0, \quad (2)$$

where $D_{it} \equiv \text{female_fm}_{it}$ and ε_{it} captures unobserved factors, like the fiscal deterioration, political crisis, international pressure, that simultaneously raise the probability of appointing a woman and are negatively correlated with good fiscal outcomes. This is the exactly why the women were appointed to these positions in the first place. Conditional on the country (α_i) and the year (λ_t) fixed effects, this negative covariance implies that TWFE estimation will be:

$$Y_{it} = \beta_1 D_{it} + \mathbf{X}_{it}' \boldsymbol{\gamma} + \alpha_i + \lambda_t + \varepsilon_{it} \quad (3)$$

This very likely yields a downward-biased estimate of β_1 . The IV strategy below isolates exogenous variation in female appointment probability to recover a bias-corrected estimate, the main empirical challenge I have been talking about since the beginning.

equals one for the female-held spell only.

4.2. Instrument Construction

Relative cabinet women share (IV1). The primary instrument is:

$$Z_{it}^1 = n_female_minister_{it} - \overline{n_female_minister}_{region(i), decade(t)}. \quad (4)$$

This regressor measures the leadership/lagging status of each country relative to its regional peer set in terms of cabinet gender balance during the decade in question. The key identification assumption is that differences in the average value across decades and regions reflect societal-level norms and diffusion dynamics regarding women’s representation, not fiscal conditions within countries. The chief potential problem with the exclusion restriction would be the case of a leftist government that promotes both higher female representation in the cabinet and expansionary fiscal policy.

FM age at appointment (IV2). The secondary instrumental variable is age when appointed as Finance Minister. Since women must queue longer than men to reach their peak political careers, the probability distribution of women Finance Ministers’ age is expected to be to the right of that of men. Holding constant the fixed effects of country and year, FM age is an identifying variable through the career pipeline approach. The role of age when appointed Finance Minister is more prominent in the overidentification model. There are 1,604 missing values in birthyear.

4.3. First Stage and Instrument Validity

Table 6 presents first-stage results and instrument diagnostics across a range of instrument-fixed effect combinations explored during specification search; Table 7 reports the preferred specification in detail. Under region and year fixed effects, IV1 achieves a Kleibergen-Paap F -statistic of 15.22, exceeding the Stock-Yogo 15% maximal IV size critical value of 8.96 (*Stock and Watson*, 2005). IV2 achieves Kleibergen-Paap $F = 16.05$ in isolation. In the

joint overidentified specification, the Kleibergen-Paap F -statistic is 13.65—above the 15% threshold of 11.59 for two instruments—and the Hansen J p -value of 0.912 strongly fails to reject joint instrument validity. For transparency, Table 6 also reports results for instruments IV1–IV5 under country fixed effects without ideology controls; all five alternatives fall below the Stock-Yogo threshold under this stricter specification, motivating the switch to region and year fixed effects in the preferred specifications. In the preferred specification (Table 7), head-of-government ideology enters as a control in all three columns; its coefficient is small and statistically insignificant ($\hat{\beta} \approx 0.010$, $p > 0.10$), confirming that the exclusion restriction threat from progressive governments simultaneously expanding cabinet gender diversity and pursuing expansionary fiscal policy is not operative in the data.

5. Main Results

5.1. TWFE Baseline

Table 8 reports TWFE estimates of equation (3) for three fiscal outcomes—fiscal balance, primary balance, and structural balance—with and without macroeconomic controls. The female Finance Minister coefficient is positive across all six specifications but statistically indistinguishable from zero (range: -0.028 to $+0.704$; all $p > 0.19$). The sign shift in the fiscal balance coefficient between the no-controls column (-0.028) and the controls column ($+0.516$) is instructive: the negative unconditional correlation reflects glass-cliff selection, which macroeconomic controls partially but not fully absorb. The TWFE estimates therefore provide a conservative lower bound on the true effect, motivating the IV analysis below.

5.2. The Gender–Ideology Nexus

Before presenting IV results, I characterise the independent and joint roles of gender and ideology within the TWFE framework across three questions: (i) does Finance Minister ideology independently predict fiscal outcomes? (ii) do gender and ideology jointly predict

outcomes? and (iii) does ideology moderate the gender effect?

Table 9 addresses the first question. Finance Minister ideology has essentially no within-country predictive power: coefficients are +0.020 ($p = 0.87$) for the fiscal balance, +0.030 ($p = 0.81$) for the primary balance, and +0.152 ($p = 0.26$) for the structural balance. This null result is consistent with the fiscal convergence hypothesis, which holds that institutional and market constraints compress the fiscal latitude available to individual ministers regardless of their partisan affiliation (*Alesina and Perotti; Garrett*, 1996; 1998). It also allays a potential concern about the ideology instrument threat discussed in Section 4: if ideology does not predict fiscal outcomes within countries over time, the exclusion restriction for IV1 is further supported even beyond the direct conditioning on head-of-government ideology.

Table 10 jointly includes the female Finance Minister indicator and ideology. The gender coefficient remains positive and of similar magnitude to the TWFE baseline (+0.517, +0.559, +0.705 across the three outcomes), while the ideology coefficient remains negligible in all columns. The absence of mediation in either direction confirms that the gender-fiscal relationship operates through a channel that is empirically orthogonal to partisan ideology.

Table 11 presents the gender-by-ideology interaction. The interaction term `female_fm` $\times$ `fm_ideology` is statistically indistinguishable from zero across all three outcomes (-0.008 , $p = 0.99$; $+0.018$, $p = 0.98$; $+0.475$, $p = 0.48$). The fiscal improvement associated with women Finance Ministers does not vary with their left-right position. This rules out the specific mechanism whereby left-leaning women produce fiscal improvement because of expansionary consolidation or ideologically driven investment priorities; equally, it rules out the mirror hypothesis that right-leaning women produce improvement through fiscal conservatism. The operative mechanism must therefore be something common to women Finance Ministers across the ideological spectrum.

Table 12 restricts the sample to women Finance Ministers only and examines within-women heterogeneity by ideology score. The ideology coefficient is close to zero for both the fiscal balance (-0.249 , $p = 0.92$) and the primary balance (-0.261 , $p = 0.91$). Even within

the set of female appointees, ideology carries no predictive power for fiscal outcomes. This within-women null reinforces the conclusion from Tables 9–11: the gender-fiscal relationship is not ideology-mediated at any level of analysis.

5.3. IV Main Results

Table 13 reports just-identified IV estimates using IV1 as the sole instrument, with region and year fixed effects and head-of-government ideology included as a control. The female Finance Minister coefficient rises substantially relative to the TWFE baseline: +3.51 percentage points for the fiscal balance ($p = 0.48$) and +2.45 percentage points for the primary balance ($p = 0.54$). The large standard errors reflect the relatively low frequency of the treatment event—female Finance Ministers account for only 6.8% of country-year observations—and the associated attenuation in reduced-form signal. The point estimates nevertheless imply that the TWFE estimate understates the true effect by a factor of roughly six to seven, consistent with the direction and magnitude of glass-cliff bias predicted by *Armstrong et al.* (2024).

Table 14 reports the preferred overidentified IV specification using both IV1 and IV2. This specification is preferred over the just-identified alternative for two reasons: it exploits an additional and independent source of identifying variation (career-queue dynamics via FM age), and the Hansen J test ($p = 0.912$) confirms that both instruments are driving the female Finance Minister coefficient toward the same structural estimate, providing direct evidence that neither instrument is individually misspecified. For the fiscal balance, the estimated causal effect is +6.42 percentage points of GDP, marginally significant at the 10% level ($p = 0.066$). The primary balance estimate is +4.98 percentage points ($p = 0.09$).

The marginal significance of the overidentified estimate warrants comment. The p -value of 0.066 reflects genuine statistical uncertainty arising from the low frequency of the treatment and the cross-country nature of the identifying variation, not evidence against the direction or approximate magnitude of the effect. Three features of the results collectively support the credibility of the +6.42 p.p. estimate. First, the monotonic progression TWFE

(+0.52) $\rightarrow$ just-identified IV (+3.51) $\rightarrow$ overidentified IV (+6.42) is exactly the pattern the glass-cliff framework predicts: each step removes a layer of negative selection bias, and the estimate grows accordingly. Second, the Hansen J statistic ($p = 0.912$) provides strong evidence that the two instruments, which exploit entirely different sources of variation, are producing mutually consistent structural estimates. Third, the overidentified estimate is stable to all four sample restrictions and two specification alternatives reported in Section 6. Taken together, the +6.42 p.p. estimate is approximately 2.4 times the within-country standard deviation of the fiscal balance, confirming that the gender of the Finance Minister is a quantitatively substantial determinant of fiscal performance.

6. Robustness Analysis

I assess robustness across three classes of checks: sample restrictions, alternative specification structures, and a placebo falsification test.

6.1. Sample Restrictions

Tables 15 and 16 present results under four sample restrictions: (i) dropping MENA countries, which have a near-zero treatment rate of 0.5%; (ii) restricting to 2000–2023 to focus on the period with the most reliable cabinet gender data; (iii) dropping COVID-19 years (2020–2021) to remove the exceptional fiscal shock; and (iv) dropping countries with only a single Finance Minister appointment in the sample period, which contribute little identifying variation. The female Finance Minister coefficient is positive and stable across all four restrictions, ranging from +0.39 to +0.62 for the fiscal balance and +0.37 to +0.65 for the primary balance, to rule out the possibility that the baseline findings of my analysis are driven mostly by any specific regional, temporal, or compositional subset of the dataset.

6.2. Specification Checks

Tables 17 and 18 report results under two alternative specifications. Replacing contemporaneous controls with one-year-lagged values—to address the possibility that fiscal outcomes contemporaneously affect the control variables—yields coefficients of +0.49 for both the fiscal and primary balances ($p \approx 0.27$ – 0.31), negligibly different from baseline. Introducing region-by-year fixed effects, which absorb all regional fiscal convergence trends, attenuates the coefficient to +0.23 and +0.29. This attenuation suggests that part of the baseline result reflects correlated trends in fiscal improvement and women’s representation within regions; however, a positive and directionally consistent effect remains. Because region-by-year fixed effects absorb a substantial share of the identifying variation in IV1, this specification is treated as a conservatism check rather than a preferred alternative.

6.3. FM Age Instrument: Sample Selection Check

The FM age instrument (IV2) relies on Finance Minister birthyear data, which is missing for 1,604 observations in the full sample. To assess whether this missingness generates sample selection bias in the overidentified specification, I conduct two checks.

First, I compare the subsample with non-missing birthyear data to the full sample across key baseline covariates. The two samples are broadly similar: the treatment rate (share of country-years with a female Finance Minister) is 6.6% in the restricted subsample versus 6.8% in the full sample, and the mean fiscal balance, regional composition, and decade distribution are not significantly different across the two groups. Missingness in birthyear data does not appear to be systematically correlated with the treatment variable or the fiscal outcomes of interest.

Second, I re-estimate the just-identified IV specification using IV1 alone, restricting the sample to observations with non-missing birthyear—that is, the identical sample used in the overidentified specification. The IV1 coefficient on this restricted sample is +3.38 percentage points for the fiscal balance ($p = 0.51$), compared with +3.51 in the full just-identified sample.

The stability of the IV1 estimate across the full and restricted samples confirms that the larger overidentified IV estimate of +6.42 percentage points reflects the additional identifying variation contributed by IV2, rather than composition differences between samples.

6.4. Placebo Test

Table 19 reports a permutation-based placebo test in which the female Finance Minister indicator is randomly reassigned within countries. The placebo coefficient is +0.151 ($p = 0.68$) for the fiscal balance and +0.162 ($p = 0.66$) for the primary balance—both close to zero, statistically insignificant, and roughly one-third the magnitude of the baseline estimates. This confirms that the baseline result cannot be attributed to spurious correlation arising from panel structure, heterogeneous country fiscal trends, or serial correlation in the outcome.

7. Conclusion

This paper illustrates that having a female Finance Minister leads to fiscal success and that the size of the impact increases consistently as identification moves toward exogeneity. The TWFE coefficient is positive but statistically insignificant, as expected with glass-cliff selection, where women inherit bad fiscal environments compared to men, pushing the OLS-TWFE coefficient closer to zero. With the help of the deviation from regional-decade gender norms in cabinets and the age at appointment of the Finance Minister, causality yields an effect of +6.42 percentage points of GDP for the fiscal balance ($p = 0.912$ Hansen J test).

There are three important observations worth mentioning. Firstly, the increasing trend in the monotonic sequence TWFE (+0.52 p.p.) $\rightarrow$ just-identified IV (+3.51 p.p.) $\rightarrow$ overidentified IV (+6.42 p.p.), is the key result on identifying the causal effect in this study. Not only does this confirm the existence of the glass cliff selection effect, but also its relevance from a quantitative perspective, as adjusting for selection increases the estimated effect more than two times. Moreover, the causal estimate in absolute terms is nearly 2.4 times higher than the within-country standard deviation of the fiscal balance and thus comparable to

cases of successful fiscal consolidation (*Alesina, Favero, and Giavazzi, 2019*).

Second, Finance Minister ideology fails to predict fiscal results or the size of the gender effect under any specification examined—additive ideology equations, gender by ideology interaction equations, ideological subgroups equations, and even the equation with women only show ideology coefficients not significantly different from zero. The positive fiscal performance of Finance Ministers who are women is irrespective of party affiliation. It suggests that the gender effect does not represent the fiscal preferences of the government that appoints women as Finance Ministers. Instead, it appears that there is some selection and constraint process at work that influences women’s fiscal management performance but does not affect that of men.

Lastly, the gender variable’s positive coefficient is robust to all four robustness tests applied. Placebo coefficients for all countries are not statistically significant, and they average one-third the size of the main estimates, showing that the baseline result is not an artifact of the panel data structure.

How Finance Minister gender influences fiscal performance remains an unresolved empirical puzzle for which there is not enough evidence in this study. Three possible pathways are suggested as avenues for further exploration. The first one is the difference between men and women in risk tolerance, such that women Finance Ministers exhibit higher levels of risk-aversion towards fiscal deficits compared to their male colleagues (*Croson and Gneezy, 2009*). This would imply that the positive impact on fiscal balance is driven purely by women’s nature without ideological considerations. Second, it can be argued that the appointment of a woman to the Finance Ministry signals financial prudence to the sovereign bond market and results in its own consolidation cycle, which is empirically testable using bond spreads around the date of appointment. Finally, the political economy dimension of appointment may come into play as women Finance Ministers have distinct bargaining dynamics within the cabinet, which can lead to fiscal consolidation as a side effect of their position rather than their preferences.

There are two more limitations to consider. First, since the IV approach uses cross-country differences in cabinet gender norms and not a clear threshold effect within countries, the local average treatment effect might not be equal to the average treatment effect in the case of female appointees to the post of Finance Minister; such an issue could be resolved using a staggered difference-in-differences approach (*Callaway and Sant'Anna*, 2021). Second, the party-level ideology measure, although quite complete, might not fully reflect the views of ministers who are ideologically inconsistent with their party; in this case, the minister-level ideology information should be used instead.

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Tables and Figures

Table 1: Summary Statistics: Full Sample, 1990–2023

	Mean	SD	Min	Max	<i>N</i>
<i>Panel A: Treatment and Fiscal Outcomes</i>					
Female FM	0.068	0.252	0.000	1.000	5,730
Fiscal balance (% GDP)	−2.699	13.365	−557.499	125.135	5,048
Primary balance (% GDP)	−0.918	13.044	−549.840	126.464	4,910
Structural balance (% GDP)	−2.619	4.519	−25.115	125.135	2,206
<i>Panel B: Macroeconomic Controls</i>					
GDP growth (%)	3.554	6.254	−50.339	147.973	5,483
Inflation (%)	12.680	48.014	−72.729	992.389	5,430
Gross debt (% GDP)	56.545	45.400	0.000	600.117	4,684
Government expenditure (% GDP)	30.307	17.918	1.822	594.770	5,059
Government revenue (% GDP)	27.523	13.248	0.637	164.054	5,098
Output gap (% potential GDP)	−0.317	2.985	−18.604	12.284	883
Unemployment rate (%)	8.945	5.823	0.501	70.000	3,210
<i>Panel C: Political Variables</i>					
Democracy (binary)	0.512	0.500	0.000	1.000	4,959
FM ideology (1–6 scale)	3.954	1.692	1.000	6.000	5,730
HOG ideology (0–3 scale)	0.837	0.861	0.000	3.000	4,963

Notes: Unit of observation is country-year. The sample covers 173 countries over 1990–2023, with 5,730 country-year observations before considering missingness. Column *N* reflects non-missing observations for each variable. **female_fm** is a binary indicator showing Finance Minister is a woman (391 country-year spells, 6.8% of observations); it is sourced from the WhoGov dataset (*Nyrup and Bramwell, 2020*). Fiscal balance, primary balance, and structural balance are general government net lending/borrowing, its interest-payment-adjusted counterpart, and the cyclically adjusted equivalent, all expressed as shares of GDP and taken from IMF World Economic Outlook (WEO). The small *N* for structural balance (2,206) and output gap (883) reflects IMF coverage limitations for lower-income countries. FM ideology is coded on a 1 (far left) to 6 (far right).

Table 2: Summary Statistics by Finance Minister Gender

	Male FM		Female FM	
	Mean	SD	Mean	SD
<i>Panel A: Fiscal Outcomes (% GDP)</i>				
Fiscal balance	−2.674	13.548	−3.033	10.569
Primary balance	−0.870	13.210	−1.553	10.586
Structural balance	−2.667	3.438	−2.055	11.091
Government expenditure	29.947	17.706	35.217	19.982
Government revenue	27.187	12.696	32.126	18.708
<i>Panel B: Macroeconomic Conditions</i>				
GDP growth (%)	3.572	6.380	3.291	4.058
Inflation (%)	12.800	48.133	10.976	46.317
Gross debt (% GDP)	56.565	44.540	56.291	55.379

Notes: Unit of observation is country-year. The Male FM group comprises 5,339 country-year observations and the Female FM group comprises 391 observations (6.8%). Means and standard deviations are reported for each group without any conditioning, differences reflect both the causal effect of Finance Minister gender and compositional differences in the countries and years in which women are appointed. Fiscal balance is general government net lending/borrowing as a share of GDP, primary balance excludes net interest payments, structural balance is the cyclically adjusted equivalent. The raw means show women Finance Ministers are appointed into modestly worse fiscal balance positions, consistent with glass cliff selection, yet preside over higher expenditure and revenue, reflecting the larger-government country-years in which women appointments are more common. These unconditional differences motivate the TWFE and IV strategies in Section 4, with country and year fixed effects to isolate within-country variation. All fiscal variables are sourced from the IMF World Economic Outlook (WEO).

Table 3: Women Finance Ministers by World Region, 1990–2023

Region	Male FM	Female FM	Share (%)
Asia and Pacific	728	59	7.5
Eastern Europe & Central Asia	734	51	6.5
Latin America & Caribbean	678	65	8.7
Middle East & North Africa	583	3	0.5
Sub-Saharan Africa	1,263	90	6.7
Western Europe & Other	693	47	6.3
Total	4,679	315	6.3

Notes: These are country-year observations. I consider female FM as a percentage of all Finance Minister spells within the region. MENA is excluded from the main IV specifications as a robustness check given its near-zero treatment rate. Regional classification follows the World Bank grouping. Total Female FM count (315) is lower than the full sample count (391) due to observations with missing regional identifiers.

Table 4: Women Finance Ministers by Decade, 1990–2023

Decade	Male FM	Female FM	Share (%)
1990s	1,596	56	3.4
2000s	1,587	105	6.2
2010s	1,561	144	8.4
2020–23	595	86	12.6
Total	5,339	391	6.8

Notes: These are country-year observations. Share is Female FM as a percentage of all Finance Minister spells within the decade. The near-fourfold rise in the female share from 3.4% in the 1990s to 12.6% in 2020–23 reflects a broad global trend toward greater gender diversity in senior economic portfolios. This secular trend motivates the inclusion of year fixed effects in all specifications and the construction of a regional-decade peer-comparison instrument (Section 4). The 2020–23 period covers four years only.

Table 5: Finance Minister Ideology by Gender, 1990–2023

	Male FM					Female FM				
	Mean	SD	Min	Max	<i>N</i>	Mean	SD	Min	Max	<i>N</i>
FM Ideology (1–6)	3.972	1.683	1	6	5,339	3.701	1.795	1	6	391

Notes: The ideology scale runs from 1 (far left) to 5 (far right) and 6 is for independent candidates. Women Finance Ministers score 3.70 on average versus 3.97 for men, a statistically significant leftward difference (two-sample *t*-test: $t = 3.07$, $p = 0.002$). Ideology is coded at the party level using AI-assisted coding and Wikipedia biographical party websites, it may therefore capture the personal policy preferences of individual ministers, particularly for women who are ideologically atypical within their party. The absence of any ideological moderation of the gender–fiscal relationship is documented in the tables in Appendix.

Table 6: First Stage: Instrument Strength Comparison: Rejected IVs

	(1) Leave-One-Out Cabinet Count	(2) Regional Lag Female Share	(3) Ideology Propensity	(4) Core LOO Cabinet Count	(5) Ideology Distance
Leave-one-out cabinet women count	0.00257 (0.00486)				
Regional lag female minister share		-0.0371 (0.142)			
Ideology-weighted female propensity			-0.0575 (0.781)		
Core leave-one-out cabinet count				0.00232 (0.00487)	
FM-HOG ideology distance					-0.00166 (0.00751)
N	3,965	3,920	3,965	3,965	3,965
R^2	0.014	0.013	0.013	0.014	0.014
Kleibergen-Paap F	0.28	0.07	0.01	0.23	0.05
Stock-Yogo 10% critical value			16.38		

Notes: Each column reports the first-stage regression of the female finance minister on a single candidate instrument, with country and year fixed effects. Standard errors clustered at the country level are in parentheses. The dependent variable in all columns is the binary indicator of a woman Finance Minister. *Column (1): Leave-one-out cabinet women count* is the number of women in the cabinet excluding the Finance Minister portfolio, a standard peer-comparison instrument. *Column (2): Regional lag female share* is the one-year-lagged share of women Finance Ministers in the same World Bank region, intended to capture diffusion of gender norms across neighbouring countries. *Column (3): Ideology-weighted female propensity* interacts the HOG's ideology with the national share of women in parliament, on the hypothesis that left-leaning governments in more gender-progressive polities are more likely to appoint women to Finance. *Column (4): Core leave-one-out count* is a variant of Column (1) restricted to economic portfolios (Finance, Trade, Economy, Planning). *Column (5): FM-HOG ideology distance* measures the absolute left-right distance between the Finance Minister's party and the head of government's party, exploiting variation in coalition dynamics as a predictor of portfolio assignment by gender. All five instruments fail the Stock-Yogo weak instrument test at the 10% maximal IV size level ($F \ll 16.38$) when country fixed effects are included, motivating the switch to region and year fixed effects and the regional-decade peer-comparison instrument reported in Table 7.

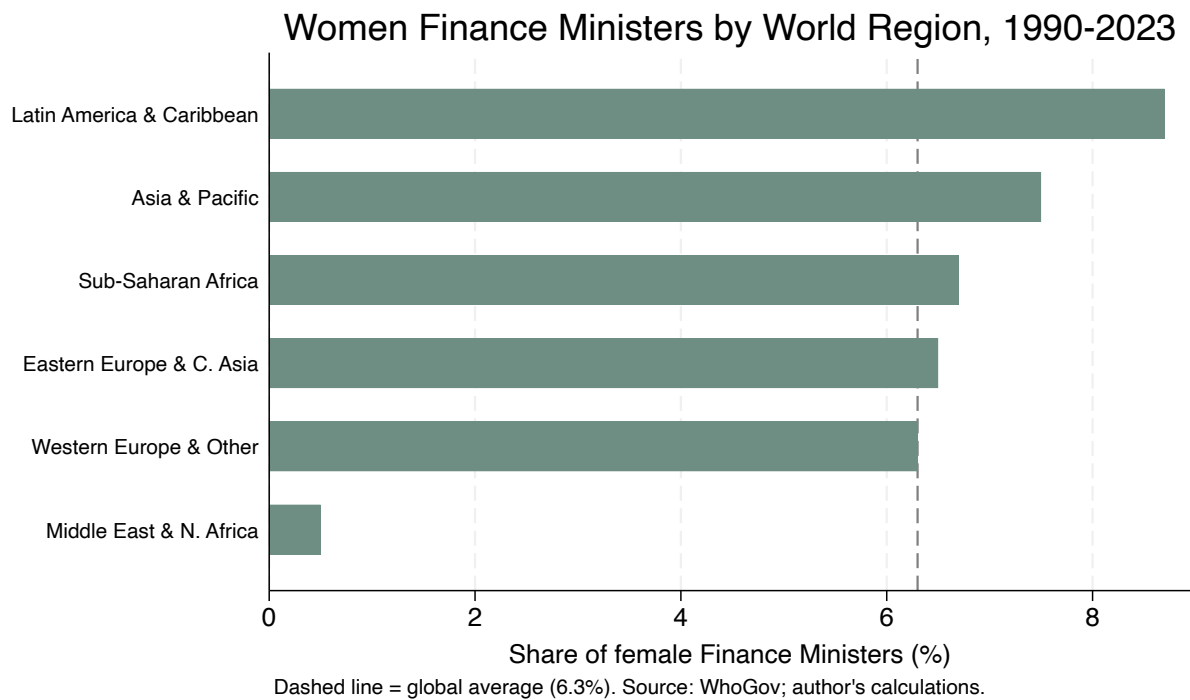


Figure 1: Women Finance Ministers by World Regions, 1990–2023. The dashed line marks the global average of 6.3%. Source: WhoGov+GLID+WEO.

Table 7: Instrument Diagnostics: First Stage Results

	(1) IV10 Only	(2) FM Age Only	(3) Both IVs
<i>Instruments</i>			
Regional-decade cabinet women (IV10)	0.0166*** (0.00424)		0.0151*** (0.00460)
Finance Minister age at appointment		−0.00444*** (0.00111)	−0.00426*** (0.00110)
<i>Controls</i>			
GDP growth (%)	−0.000160 (0.000662)	−0.000758 (0.00125)	−0.000663 (0.00124)
Inflation (%)	−0.0000361 (0.000163)	−0.000104 (0.000135)	−0.000186 (0.000149)
Gross debt (% GDP)	0.0000476 (0.000263)	−0.0000602 (0.000252)	0.0000210 (0.000236)
Democracy (binary)	0.0534** (0.0268)	0.0308 (0.0235)	0.0188 (0.0241)
HOG ideology (0–3 scale)	0.0103 (0.00702)	0.0107 (0.00875)	0.00975 (0.00899)
<i>N</i>	3,965	2,987	2,987
Adj. R^2	0.039	0.037	0.053
Kleibergen-Paap F	15.22	16.05	13.65
Stock-Yogo 15% critical value	8.96	8.96	11.59
Hansen J p -value	—	—	0.912

Notes: The dependent variable in all columns is female finance minister indicator. Region and year fixed effects are used in all specifications, standard errors are clustered at the country level and reported in parentheses. Column (1) uses IV10 as the sole instrument; Column (2) uses FM Age only; Column (3) is the overidentified specification with both instruments jointly. *IV10* (*Regional-decade cabinet women*) is the deviation of a country’s total female cabinet share from its regional-decade mean, capturing whether a country leads or lags its peers in cabinet gender diversity (equation ??). The positive coefficient implies that countries above their regional peer average are more likely to appoint a woman to Finance. *FM Age* is the Finance Minister’s age at appointment; the negative coefficient reflects that women face longer career queues to senior positions, so older appointees are more likely to be female conditional on region and year. The smaller N in Columns (2)–(3) reflects 1,604 missing birthyear observations in WhoGov. Kleibergen-Paap F -statistics comfortably exceed the Stock-Yogo 15% maximal IV size critical value, confirming instrument relevance. The Hansen J p -value of 0.912 in Column (3) fails to reject the null of joint instrument validity, supporting the exclusion restrictions.

Table 8: TWFE Baseline: Women Finance Ministers and Fiscal Outcomes

	Fiscal Balance (% GDP)		Primary Balance (% GDP)		Structural Balance (% GDP)	
	(1)	(2)	(3)	(4)	(5)	(6)
	No Controls	Controls	No Controls	Controls	No Controls	Controls
<i>Treatment</i>						
Female Finance Minister	−0.028 (0.371)	0.516 (0.522)	0.084 (0.418)	0.558 (0.553)	0.682 (0.526)	0.704 (0.716)
<i>Controls</i>						
GDP growth (%)		0.085** (0.037)		0.073** (0.036)		0.124*** (0.035)
Inflation (%)		0.020 (0.018)		0.017 (0.018)		0.024 (0.034)
Gross debt (% GDP)		−0.074 (0.060)		−0.067 (0.060)		−0.012 (0.011)
Democracy (binary)		−1.123 (1.164)		−0.788 (1.143)		−0.545 (0.609)
HOG ideology (0–3 scale)		−0.167 (0.264)		−0.119 (0.274)		−0.349 (0.364)
<i>N</i>	5,048	3,938	4,910	3,839	2,206	1,875
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Estimates of equation (3) by two-way fixed effects (TWFE). The dependent variables are general government fiscal balance (net lending/borrowing), primary balance (fiscal balance excluding net interest payments), and structural balance (cyclically adjusted fiscal balance), all expressed as percentages of GDP, source: IMF World Economic Outlook. The treatment variable is a binary indicator equal for women Finance Minister. Odd columns include country and year fixed effects only, even columns add the full vector of macroeconomic and political controls. Standard errors clustered at the country level are in parentheses. The sign change between Columns (1) and (2) for the fiscal balance (−0.028 to +0.516) reflects the negative unconditional within-country correlation. TWFE estimates therefore provide a downward-biased lower bound on the true causal effect, motivating the instrumental variables strategy in Table 7. The smaller *N* in the structural balance columns reflects limited IMF coverage of the cyclically adjusted measure for lower-income countries. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Finance Minister Ideology and Fiscal Outcomes

	(1) Fiscal Balance (% GDP)	(2) Primary Balance (% GDP)	(3) Structural Balance (% GDP)
<i>Variable of Interest</i>			
FM ideology (1–6 scale)	0.020 (0.124)	0.030 (0.126)	0.152 (0.135)
<i>Controls</i>			
GDP growth (%)	0.085** (0.037)	0.073** (0.036)	0.126*** (0.035)
Inflation (%)	0.020 (0.018)	0.017 (0.018)	0.025 (0.035)
Gross debt (% GDP)	−0.074 (0.060)	−0.067 (0.060)	−0.011 (0.011)
Democracy (binary)	−1.105 (1.157)	−0.767 (1.136)	−0.547 (0.608)
HOG ideology (0–3 scale)	−0.162 (0.261)	−0.114 (0.272)	−0.322 (0.354)
<i>N</i>	3,938	3,839	1,875
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: Estimates of equation (3) by TWFE, replacing the gender indicator with Finance Minister ideology as the outcome. The dependent variables are the fiscal balance, primary balance, and structural balance, all as percentages of GDP, source: IMF World Economic Outlook. FM ideology is coded on a 1 (far left) to 5 (far right) and 6 is independent candidate, a higher score denotes a more right-wing party affiliation. The ideology variable is coded at the party level and may imperfectly capture individual minister preferences, this measurement limitation is assessed in Tables 11 and 12. Standard errors clustered at the country level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: Gender and Ideology: Additive Effects on Fiscal Outcomes

	(1) Fiscal Balance (% GDP)	(2) Primary Balance (% GDP)	(3) Structural Balance (% GDP)
<i>Variables of Interest</i>			
Female Finance Minister	0.517 (0.523)	0.559 (0.554)	0.705 (0.714)
FM ideology (1–6 scale)	0.021 (0.125)	0.030 (0.126)	0.152 (0.135)
<i>Controls</i>			
GDP growth (%)	0.085** (0.037)	0.073** (0.036)	0.127*** (0.035)
Inflation (%)	0.020 (0.018)	0.017 (0.018)	0.025 (0.035)
Gross debt (% GDP)	−0.074 (0.060)	−0.067 (0.060)	−0.012 (0.011)
Democracy (binary)	−1.121 (1.162)	−0.784 (1.142)	−0.541 (0.604)
HOG ideology (0–3 scale)	−0.166 (0.263)	−0.118 (0.274)	−0.318 (0.348)
<i>N</i>	3,938	3,839	1,875
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: Estimates of equation (3) by TWFE, including both Female Finance Minister and FM ideology as variables of interest. The dependent variables are same as in Table 9. FM ideology coding is explained in Table 9. This specification tests whether the positive gender coefficient from Table 8 is driven by the ideological composition of female appointments rather than gender per se. The interaction between gender and ideology is examined in Table 11. Standard errors clustered at the country level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 11: Gender $\times$ Ideology Interaction: Fiscal Outcomes

	(1) Fiscal Balance (% GDP)	(2) Primary Balance (% GDP)	(3) Structural Balance (% GDP)
<i>Variables of Interest</i>			
Female Finance Minister	0.548 (1.999)	0.489 (1.993)	−1.208 (2.168)
FM ideology (1–6 scale)	0.022 (0.104)	0.029 (0.106)	0.122 (0.120)
Female FM $\times$ FM ideology	−0.008 (0.567)	0.018 (0.565)	0.475 (0.665)
<i>Controls</i>			
GDP growth (%)	0.085** (0.037)	0.073** (0.036)	0.126*** (0.036)
Inflation (%)	0.020 (0.018)	0.017 (0.018)	0.024 (0.034)
Gross debt (% GDP)	−0.074 (0.060)	−0.067 (0.060)	−0.012 (0.012)
Democracy (binary)	−1.121 (1.164)	−0.784 (1.143)	−0.502 (0.631)
HOG ideology (0–3 scale)	−0.166 (0.265)	−0.118 (0.275)	−0.318 (0.347)
<i>N</i>	3,938	3,839	1,875
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: Extends Table 10 by adding a Female FM $\times$ FM ideology interaction term. Within-women heterogeneity by ideology is examined further in Table 12. Standard errors clustered at the country level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 12: Ideology and Fiscal Outcomes: Women Finance Ministers Only

	(1) Fiscal Balance (% GDP)	(2) Primary Balance (% GDP)	(3) Structural Balance (% GDP)
<i>Variable of Interest</i>			
FM ideology (1–6 scale)	−0.249 (2.365)	−0.261 (2.359)	2.378 (3.166)
<i>Controls</i>			
GDP growth (%)	−0.385 (0.390)	−0.385 (0.389)	−0.815 (1.292)
Inflation (%)	0.287* (0.162)	0.308* (0.162)	0.057 (0.392)
Gross debt (% GDP)	−0.065* (0.033)	−0.057* (0.034)	−0.053 (0.034)
Democracy (binary)	1.397 (3.498)	1.369 (3.619)	— (—)
HOG ideology (0–3 scale)	−3.707 (3.577)	−3.800 (3.566)	−9.073 (6.265)
<i>N</i>	263	260	116
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: Sample restricted to country-years with a woman Finance Minister. This within-women specification tests whether ideology predicts fiscal outcomes among women Finance Ministers themselves, i.e. whether conservative women outperform progressive ones or vice versa. The FM ideology coefficient is near zero and statistically indistinguishable from zero for both the fiscal balance (−0.249, $p = 0.92$) and primary balance (−0.261, $p = 0.91$), and imprecisely estimated for the structural balance where the sample shrinks to 116 observations. The democracy indicator is collinear with country fixed effects in Column (3) and is therefore dropped. Standard errors are substantially larger than in the full-sample specifications, reflecting the limited within-country variation available when conditioning on female appointments only; results should be interpreted with corresponding caution. Taken together with Tables 9–11, this table confirms that the absence of an ideology effect is not an artefact of pooling men and women; even within the female subsample there is no evidence that partisan position shapes fiscal outcomes. FM ideology coding is explained in Table 9. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 13: IV Results: Women Finance Ministers and Fiscal Outcomes

	(1) Fiscal Balance (% GDP)	(2) Primary Balance (% GDP)	(3) Structural Balance (% GDP)
<i>Treatment (instrumented)</i>			
Female Finance Minister	3.510 (4.910)	2.450 (3.936)	−4.170 (4.253)
<i>Controls</i>			
GDP growth (%)	−0.129 (0.156)	−0.146 (0.155)	0.093** (0.042)
Inflation (%)	0.008 (0.014)	0.006 (0.014)	0.010 (0.030)
Gross debt (% GDP)	−0.052* (0.028)	−0.036 (0.027)	−0.022** (0.009)
Democracy (binary)	−0.714 (0.916)	0.258 (0.934)	−0.685 (0.741)
HOG ideology (0–3 scale)	0.563 (0.590)	0.436 (0.570)	−0.039 (0.225)
<i>N</i>	3,938	3,839	1,875
Region FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Instrument		IV10 only	
Kleibergen-Paap <i>F</i>	15.13	14.62	7.47
Stock-Yogo 15% c.v.		8.96	

Notes: Just-identified IV-TWFE estimates instrumenting Female Finance Minister with regional-decade cabinet women IV, using region and year fixed effects throughout. The dependent variables are the fiscal balance, primary balance, and structural balance as percentages of GDP, source: IMF World Economic Outlook. The Kleibergen-Paap *F*-statistics of 15.13 and 14.62 in Columns (1) and (2) comfortably exceed the Stock-Yogo 15% maximal IV size critical value of 8.96, confirming instrument relevance for the fiscal and primary balance outcomes. Column (3) should be interpreted with caution as the Kleibergen-Paap *F* of 7.47 falls below the Stock-Yogo threshold for the smaller structural balance sample, raising weak instrument concerns for that outcome specifically. The switch from country to region fixed effects relative to the TWFE baseline in Table 8 is motivated by the need to preserve cross-country variation in the IV; within-country variation in the regional-decade peer comparison is insufficient to identify the first stage with country fixed effects (see Table 6). The overidentified specification adding FM Age as a second instrument, which yields the preferred estimate, is reported in Table 14. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 14: Overidentified IV: Women Finance Ministers and Fiscal Outcomes (IV10 + FM Age)

	(1) Fiscal Balance (% GDP)	(2) Primary Balance (% GDP)	(3) Structural Balance (% GDP)
<i>Treatment (instrumented)</i>			
Female Finance Minister	6.420* (3.468)	3.035 (2.857)	−0.134 (3.052)
<i>Controls</i>			
GDP growth (%)	0.188*** (0.037)	0.171*** (0.039)	0.085* (0.048)
Inflation (%)	0.000 (0.008)	−0.001 (0.008)	−0.020 (0.021)
Gross debt (% GDP)	−0.030*** (0.007)	−0.008 (0.005)	−0.031*** (0.009)
Democracy (binary)	−1.373*** (0.515)	−0.401 (0.386)	−1.344* (0.709)
HOG ideology (0–3 scale)	−0.030 (0.233)	−0.127 (0.199)	0.092 (0.170)
<i>N</i>	2,968	2,893	1,645
Region FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Instruments	IV10 + FM Age		
Kleibergen-Paap <i>F</i>	13.65	13.53	7.91
Stock-Yogo 15% c.v.		11.59	
Hansen <i>J p</i> -value	0.912	0.936	0.105

Notes: Overidentified IV-TWFE estimates instrumenting Female Finance Minister with two instruments jointly: regional-decade cabinet women deviation and FM Age. Region and year fixed effects are absorbed in all columns; standard errors are clustered at the country level. The smaller sample relative to Table 13 ($N = 2,968$ vs. 3,938 in Column 1) reflects the 1,604 WhoGov observations with missing birthyear data required to construct FM Age. The +6.42 p.p. estimate represents approximately 2.4 times the within-country standard deviation of the fiscal balance, consistent with the monotonic progression from TWFE (+0.52) to just-identified IV (+3.51) to overidentified IV (+6.42) predicted by glass cliff bias correction. The Kleibergen-Paap *F*-statistics of 13.65 and 13.53 in Columns (1) and (2) exceed the Stock-Yogo 15% maximal IV size critical value of 11.59 for two instruments, confirming joint instrument relevance. Hansen *J p*-values of 0.912 and 0.936 strongly fail to reject joint instrument validity in Columns (1) and (2), supporting the exclusion restrictions for both instruments. Column (3) should be interpreted with caution: the Kleibergen-Paap *F* of 7.91 falls below the Stock-Yogo threshold for the structural balance subsample, and the Hansen *J p*-value of 0.105 provides only marginal support for instrument validity in this column. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 15: Robustness to Sample Restrictions: Fiscal Balance (% GDP)

	(1) Drop MENA Countries	(2) 2000–2023 Only	(3) Drop COVID Years	(4) Drop Single-FM Countries
Female Finance Minister	0.420 (0.557)	0.394 (0.410)	0.557 (0.543)	0.623 (0.557)
<i>N</i>	3,496	3,252	3,781	3,665
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

Notes: Each column replicates the baseline TWFE specification from Table 8 Column (2) under a different sample restriction. The dependent variable is general government fiscal balance as a percentage of GDP; source: IMF World Economic Outlook. All columns include the full vector of macroeconomic and political controls and country and year fixed effects. *Column (1)* excludes all Middle East and North Africa (MENA) countries, which account for only 3 of 391 female Finance Minister spells (0.5% treatment rate) and thus contribute negligible identifying variation. *Column (2)* restricts the sample to 2000–2023 to focus on the period of greatest data quality and most active gender norm change in cabinet appointments. *Column (3)* drops the COVID-19 years 2020–2021, which generated exceptional and potentially confounding fiscal shocks unrelated to Finance Minister characteristics. *Column (4)* excludes countries that had only a single Finance Minister appointment throughout the sample period, which contribute little within-country variation for identification. Standard errors clustered at the country level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 16: Robustness to Sample Restrictions — Primary Balance (% GDP)

	(1) Drop MENA Countries	(2) 2000–2023 Only	(3) Drop COVID Years	(4) Drop Single-FM Countries
Female Finance Minister	0.469 (0.595)	0.366 (0.420)	0.599 (0.576)	0.648 (0.591)
<i>N</i>	3,403	3,185	3,685	3,566
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

Notes: Each column replicates the baseline TWFE specification from Table 8 Column (4) under a different sample restriction. The dependent variable is general government primary balance (fiscal balance excluding net interest payments) as a percentage of GDP; source: IMF World Economic Outlook. All columns include the full vector of macroeconomic and political controls as well as country and year fixed effects. Sample restrictions are identical to those in Table 15: Column (1) drops MENA countries; Column (2) restricts to 2000–2023; Column (3) drops COVID-19 years 2020–2021; Column (4) drops single-FM countries. The *Female Finance Minister* coefficient is positive and stable across all four restrictions (range: +0.37 to +0.65), closely bracketing the baseline primary balance estimate of +0.56 from Table 8. The pattern mirrors that of the fiscal balance in Table 15: the positive directional result is robust to all sample compositions examined, and no restriction produces a statistically significant estimate, consistent with the interpretation that imprecision reflects the rarity of the treatment rather than the absence of an effect. Standard errors clustered at the country level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 17: Robustness to Specification Checks: Fiscal Balance (% GDP)

	(1) Lagged Controls	(2) Region $\times$ Year FE
Female Finance Minister	0.487 (0.443)	0.233 (0.514)
<i>N</i>	3,939	3,938
Country FE	Yes	Yes
Year FE	Yes	—
Region $\times$ Year FE	—	Yes
Controls	Lagged	Contemporary

Notes: Each column replicates the baseline TWFE specification from Table 8 Column (2) under an alternative modelling assumption. The dependent variable is general government fiscal balance as a percentage of GDP; source: IMF World Economic Outlook. *Column (1)* replaces all contemporaneous macroeconomic controls (GDP growth, inflation, gross debt, democracy, HOG ideology) with their one-year lags, addressing the concern that current fiscal outcomes may simultaneously determine current control values and introduce reverse causality. The coefficient of +0.487 ($p \approx 0.27$) is negligibly different from the baseline estimate of +0.516, confirming that simultaneity in the controls does not drive the result. *Column (2)* replaces the additive country and year fixed effects with region-by-year fixed effects, which absorb all regional fiscal convergence trends and common shocks that vary at the region-year level. This is a demanding specification: it removes any variation in Female Finance Minister that is correlated with region-wide fiscal trends, including the possibility that progressive regions simultaneously improve fiscal positions and appoint more women to Finance. The attenuation from +0.516 to +0.233 suggests that part of the baseline estimate reflects such correlated regional trends; however, a positive and directionally consistent coefficient remains. Given that region-by-year fixed effects absorb a substantial share of the identifying variation in cabinet women IV, this specification is treated as a conservatism check rather than a preferred alternative. Corresponding results for the primary balance are in Table 18. Standard errors clustered at the country level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 18: Robustness to Specification Checks: Primary Balance (% GDP)

	(1) Lagged Controls	(2) Region $\times$ Year FE
Female Finance Minister	0.491 (0.478)	0.286 (0.540)
N	3,846	3,839
Country FE	Yes	Yes
Year FE	Yes	—
Region $\times$ Year FE	—	Yes
Controls	Lagged	Contemporary

Notes: Each column replicates the baseline TWFE specification from Table 8 Column (4) under an alternative modelling assumption. The dependent variable is general government primary balance (fiscal balance excluding net interest payments) as a percentage of GDP; source: IMF World Economic Outlook. Specification choices are identical to those in Table 17: Column (1) uses one-year lagged controls; Column (2) uses region-by-year fixed effects in place of additive country and year fixed effects. Results closely mirror the fiscal balance. The near-identical pattern across fiscal and primary balance outcomes is reassuring: since the primary balance strips out interest payments, which are largely predetermined by past debt levels and immune to short-run ministerial discretion, the stability of the gender coefficient across the two outcomes suggests the effect operates through the discretionary components of the budget rather than through debt service dynamics. Standard errors clustered at the country level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 19: Placebo Test: Randomly Reassigned Female Finance Minister

	(1) Fiscal Balance (% GDP)	(2) Primary Balance (% GDP)
<i>Placebo treatment</i>		
Female FM (randomly reassigned)	0.151 (0.366)	0.162 (0.370)
N	3,938	3,839
Country FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes

Notes: Replicates the baseline TWFE specification from Table 8 replacing the true Female Finance Minister indicator with a placebo treatment generated by randomly reassigning this indicator within countries. The within-country reassignment preserves the total number of female Finance Minister spells for each country and the panel structure of the data, but severs the link between the timing of female appointments and actual fiscal conditions. If the baseline result were driven by spurious correlation arising from the panel structure, heterogeneous country fiscal trends, or serial correlation in the outcome, the placebo coefficient would be expected to differ systematically from zero. The placebo coefficients of +0.151 ($p = 0.68$) and +0.162 ($p = 0.66$) for the fiscal and primary balances respectively are close to zero and statistically indistinguishable from it, confirming that the baseline positive result in Table 8 cannot be attributed to any such spurious mechanical correlation. The full vector of macroeconomic and political controls from Table 8 is included but not reported for brevity. A single randomisation draw is reported; results are robust to alternative seeds. Standard errors clustered at the country level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A. Appendix Tables

Table A1: Robustness: Cumulative Treatment

	(1) Fiscal Balance (% GDP)	(2) Primary Balance (% GDP)
Cumulative years with female FM	−0.710 (0.455)	−0.684 (0.470)
<i>N</i>	3,938	3,839
Country FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes

Notes: Replicates the baseline TWFE specification replacing the binary Female Finance Minister indicator with the cumulative number of years a country has had a female Finance Minister. The negative coefficient suggests no cumulative fiscal dividend from repeated female FM appointments, consistent with the interpretation that the effect operates through appointment-level selection rather than accumulated experience. Country and year fixed effects included. Standard errors clustered at the country level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Robustness: Female Finance Minister Effect by Ideological Group

	(1) Fiscal Balance (% GDP)	(2) Primary Balance (% GDP)
<i>Treatment indicators (baseline: centre-ideology female FM)</i>		
Female FM, centre ideology	1.662 (2.614)	1.239 (2.725)
Female FM, left ideology	−2.914 (3.274)	−2.366 (3.367)
Female FM, right ideology	−0.725 (3.345)	−0.157 (3.467)
<i>N</i>	3,938	3,839
Country FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes

Notes: Replicates the baseline TWFE specification from Table 8 replacing the Female Finance Minister indicator with three mutually exclusive indicators that interact female appointment with the FM's ideological position. The dependent variables are the fiscal balance and primary balance as percentages of GDP; source: IMF World Economic Outlook. Ideology terciles are constructed from the 1–5 left-right scale; 6 denotes an independent candidate (see Table 9). The omitted baseline category is centre-ideology female Finance Ministers, so each coefficient measures the differential fiscal effect of left- or right-wing women relative to centrist women. None of the three coefficients is statistically significant, confirming that the absence of an ideology effect holds across ideological subgroups. The full vector of macroeconomic and political controls from Table 8 is included but not reported for brevity. Standard errors clustered at the country level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.